RiskLens Grading Rubric

RiskLens Criterion Set v2.0 | Build Custom AI

This file defines all 24 assessment criteria for the RiskLens course project. Module rubrics reproduce the same standards. If wording differs, this consolidated rubric governs criteria; published course policy governs administrative matters.

How grading works

Each criterion is Met or Not Yet. A competency is Mastered only when every criterion in that group is Met. The course project is Mastered when all five competencies are Mastered. There are no weighted modules, points, averages, deductions or bonus credit. Model evaluation scores remain technical evidence, not course grades.

Submit one evolving notebook link at each module milestone, with saved outputs, a short completion summary and known issues. Keep a dated snapshot of each submission. Modules 1-5 complete C1-C5 respectively; there are 24 criteria in total.

Module	Competency	Criteria	Notebook evidence
1	Visual Inspection and Evaluation	C1.1-C1.5	D1.1-D1.7
2	Grounded Retrieval and Generation	C2.1-C2.5	D2.1-D2.7
3	Model Adaptation and Evaluation	C3.1-C3.5	D3.1-D3.7
4	Model Optimization and Deployment Decisions	C4.1-C4.5	D4.1-D4.4
5	Integrated System Delivery and Defense	C5.1-C5.4	D4.5-D4.7 plus final delivery

MUST PASS review order

Review the MUST PASS criterion first for each module. If it is Not Yet, return that submission with targeted feedback before reviewing the remaining criteria. Every other criterion is also required; MUST PASS determines review order, not extra weight.

Module 1: C1.1 frozen split. Module 2: C2.3 grounded answers. Module 3: C3.2 evaluation separation. Module 4: C4.2 predeclared gate. Module 5: C5.1 real integration.

Evidence, feedback and revision

One continuing notebook | Five module milestones

Evidence that can be assessed

Assess saved outputs and linked artifacts; graders do not need to repeat model training. A missing evaluation output does not demonstrate a result. Preserve real prompts, configurations, data versions and outputs as the work happens. Dated snapshots or version history support freezing; cell position alone does not prove chronology.

Returned submissions and integrity

Return an inaccessible or incomplete submission for correction before assessment. Missing required outputs, a broken run/load path or exposed credentials must be resolved. Refer fabricated measurements, reconstructed evidence or copied work for academic review under course policy; do not convert these into point deductions.

Reassessment

Feedback identifies each Not Yet criterion, the missing or incorrect evidence and the required revision. Resubmit a short revision plan and the revised evidence. Reassess Not Yet criteria and any previously Met evidence affected by the changes. Attempt limits, waiting periods, enrolment rules and formal grade codes follow the published course policy.

Optional extensions

GraphRAG/KAG implementations, independent distillation, multi-parent merging, multi-GPU training, GGUF serving, a hosted demo and extra product categories are optional extensions. They do not compensate for a Not Yet criterion. No hidden-holdout score, timed exam or recorded-defense rule is introduced by this rubric.

Technical scores are not course grades. Defect F1, recall, format rates, judge scores, win-rates and quality-drop thresholds remain part of the evidence. A Met judgment depends on the stated criterion, not on an average across these metrics.

C1 | Visual Inspection and Evaluation

Module 1 - Visual Defect Inspector | Criteria 1/2

Competency: Build and evaluate a vision-language inspector that diagnoses and corrects the inherited baseline defects, returns reliable structured outputs, improves defect F1 on a frozen test set, and justifies prompting and image-resolution choices using measured quality and cost.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C1.1 Your evaluation split is frozen MUST PASS	D1.1 SPLIT cell; saved seed/counts; pre-iteration notebook snapshot or version history.	Before prompt iteration, freeze at least 40 test images with at least 20 per class and a separate development set. Print seed, membership identifiers and class counts. Preserve a dated snapshot or version record. Use development data for choices and disclose any later test exposure.	Changing the split after seeing results; overlapping development and test images; treating cell position alone as proof of timing.
C1.2 Your inspector fixes the inherited defects	D1.2 bug diagnoses; D1.3 inspect(); representative outputs and fallback checks.	Diagnose and fix the negation-unsafe parser, context-free prompt and empty defect description. Each diagnosis links a real symptom to its cause and fix. inspect() returns the required JSON fields and valid verdict, uses a domain prompt and structured output, and handles malformed model replies without an uncaught parsing error. Justify the VLM choice.	Leaving one bug unresolved; treating "no defects" as defective; returning empty defect evidence; a parser failure that crashes the pipeline.
C1.3 Your prompt iteration is measured	D1.3-D1.4 prompt ladder; saved development outputs.	Compare at least three prompt iterations on the same development set, including a few-shot attempt. Keep every prompt, development accuracy, reason for the change and final selection visible. Include the few-shot token-cost observation. Select the final prompt before final-test scoring.	Only showing the winning prompt; unmeasured prompt changes; choosing a prompt from test-set results; omitting the few-shot attempt.

Keep the evidence locations, saved outputs and version records visible in the submitted notebook. The same criteria appear in the module rubric.

C1 | Visual Inspection and Evaluation

Module 1 - Visual Defect Inspector | Criteria 2/2

All criteria in this competency must be Met. C1.1 is the MUST PASS review gate.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C1.4 Your performance and error claims are supported	D1.5 images and analysis; D1.6 saved Stage 1 evaluation and ledger.	On the frozen test set, report baseline and final defect F1 and the confusion matrix; final F1 exceeds inspect_v0. Analyze at least three actual errors with plausible hypotheses. If fewer exist, analyze all errors and enough clearly labeled borderline cases to total three. Save the evaluation outputs and explain the baseline recall inflation.	Comparing different test sets; substituting accuracy or recall for defect F1; inventing errors; reporting numbers without saved outputs.
C1.5 Your resource recommendation follows the evidence	D1.6 timing/cost; D1.7 resolution table and recommendation.	Measure development accuracy and seconds per image at three or more image sizes under stated hardware and settings. Record estimated cost per 1,000 inspections with the assumed hourly price. Recommend an image size based on the observed quality/time trade-off and explain what the processor actually does with the resized inputs.	Only reporting memory; comparing different image sets; claiming higher input resolution always helps; omitting price or runtime assumptions.

Freeze the split before experimentation. If reassessment exposes the test set, record that exposure and agree a new evaluation version before further tuning. Do not manufacture three errors when fewer exist.

C2 | Grounded Retrieval and Generation

Module 2 - RAG Evidence Engine | Criteria 1/2

Competency: Build and evaluate a retrieval-augmented evidence engine that finds relevant product information, supports its answers with auditable citations, refuses unsupported requests, and justifies retrieval and graph-based design choices using controlled measurements.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C2.1 Your corpus and gold questions are valid	D2.1 CORPUS; D2.2 GOLD; source provenance and frozen snapshot.	Include at least 100 real reviews, all four official documents verbatim with their IDs, and at least two authored documents containing testable facts and a cross-document link. Before indexing, freeze at least 20 corpus-based gold questions, including three multi-hop cases, with required source IDs and reference answers. Preserve the version record.	Invented reviews presented as real; altered official facts; gold questions chosen after retrieval trials; multi-hop labels on single-source questions.
C2.2 Your retriever improves under a fair comparison	D2.3 retrieve(); D2.6 evaluation; D2.7 upgrade experiment.	Beat retrieve_v0 on recall@4 using the same corpus, frozen gold questions and metric implementation. Save baseline/final recall at k = 1, 2, 4 and 8. Measure one upgrade beyond the query prefix on unchanged questions, show before/after results, and justify adopting or rejecting it. A rejected upgrade is valid evidence.	Changing the gold set or corpus between arms; claiming recall@8 beats baseline recall@4; reporting only the prefix fix; omitting a negative experiment.
C2.3 Your answers are grounded and refuse unsupported claims MUST PASS	D2.4 rag_answer(); answer/hit pairs; five-question adversarial set and saved responses.	Return the answer and the actual retrieved hits. Citations resolve to those source IDs and support the associated claims. Preselect five genuinely unanswerable questions; at least four receive a refusal containing a REFUSAL_PHRASES phrase without an invented answer. Report all five outcomes; separately report any additional adversarial cases.	Decorative or nonexistent citations; refusing and then inventing an answer; changing adversarial questions after seeing failures; counting a phrase match as sufficient proof.

Keep the evidence locations, saved outputs and version records visible in the submitted notebook. The same criteria appear in the module rubric.

C2 | Grounded Retrieval and Generation

Module 2 - RAG Evidence Engine | Criteria 2/2

All criteria in this competency must be Met. C2.3 is the MUST PASS review gate.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C2.4 Your generation evaluation measures both axes	D2.5 scoring outputs; D2.6 evaluation and ledger; judge protocol.	Score faithfulness and answer relevance separately over the frozen gold set. State the judge or evaluation tool, scoring instructions, sample size and aggregation, and preserve representative answer-level results. Explain judge bias, especially shared generator/judge weights. Save the Stage 2 outputs and ledger; reference-substring matches alone do not establish faithfulness.	Only measuring retrieval; using answer-substring matches as a faithfulness score; unnamed judge; no caveat about self-evaluation or judgment limits.
C2.5 Your retrieval design choice is justified	Design note beside D2.7; one question and its linked source IDs.	Trace a question requiring linked facts across documents. Explain whether graph-assisted retrieval would help and identify its extraction, indexing and maintenance costs. Distinguish the notebook graph-hop illustration from full GraphRAG and KAG systems. Justify the retrieval approach used in RiskLens; implementing a complete graph framework is not required.	Calling any vector index GraphRAG; presenting a toy graph as a complete KAG system; claiming graphs improve every query; no concrete linked question.

The starter counts a gold question as a retrieval hit only when every required source is retrieved. Retain that all-required-sources definition for baseline/final comparisons and label it clearly. Citation support and refusal meaning still require inspection beyond phrase matching.

C3 | Model Adaptation and Evaluation

Module 3 - Fine-Tuned Analyst | Criteria 1/2

Competency: Curate and document teacher-generated training data, train and reload a QLoRA Analyst, and evaluate it against the untuned base model on separated data using format, content and pairwise comparisons, with a controlled ablation and honest limitations.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C3.1 Your training data is real and curated	D3.1 teacher calls and examples; D3.2 DATASET_CARD and cleaning counts.	Generate at least 150 raw pairs with a named real teacher model and varied inputs using make_input(). Document the generation strategy. Perform deduplication and train/test decontamination, printing before/after and removal counts. Complete every dataset-card field and report retained training size. Template-only generated reports do not meet the requirement.	Using the practice template as teacher output; claiming 150 retained examples when only raw count was measured; describing cleaning without running it; incomplete card.
C3.2 Your evaluation data is separated from training MUST PASS	D3.3 analyst_test_rows; similarity calculation; dataset-card snapshot before training.	Freeze at least 25 Analyst test rows before training. Drop contaminated training rows and show the maximum train-to-test cosine similarity is at most 0.90 using a declared embedding model and method. Preserve the frozen version and counts. Explain that the similarity screen is a leakage heuristic, not proof of independence.	Training on evaluation examples; removing inconvenient test rows; reporting the mean instead of maximum similarity; using cell order as the only chronology evidence.
C3.3 Your QLoRA adapter is trained and reloadable	D3.4 training configuration, loss outputs, adapter artifact and reload demonstration.	Complete QLoRA training with a configuration appropriate to the runtime. Preserve base model ID/revision, adapter settings and training parameters, and interpret the observed loss. Save the learner-trained adapter to an accessible path or Hub location and demonstrate loading it. The delivery path loads saved weights instead of retraining on every Run all.	A configuration with no completed training; an adapter path that is inaccessible; loading only the base; no loss interpretation; retraining unexpectedly during delivery.

Keep the evidence locations, saved outputs and version records visible in the submitted notebook. The same criteria appear in the module rubric.

C3 | Model Adaptation and Evaluation

Module 3 - Fine-Tuned Analyst | Criteria 2/2

All criteria in this competency must be Met. C3.2 is the MUST PASS review gate.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C3.4 Your base and tuned comparison is valid	D3.5 generations and judgments; D3.6 WEEK3_RESULTS, Stage 3 output and ledger.	Evaluate genuinely untuned base and tuned outputs on the same frozen rows using format adherence, content rubric scores and blind randomized A/B judgments. Name a separate, stronger judge and state its limitations. Show wins, losses, ties and denominator. Populate WEEK3_RESULTS from saved evidence. No minimum win-rate or tuning-improvement margin is imposed.	The base still has an active adapter; fixed A/B position; a percentage without counts; judge equals generator without disclosure; scores copied without supporting outputs.
C3.5 Your ablation supports a defensible conclusion	D3.7 two adapter runs, configuration difference, comparison and conclusion.	Change exactly one training variable: rank, epochs, learning rate or training-data size. Train a second adapter and compare both runs on the same frozen rows with the same evaluation protocol. Preserve both results. Explain what changed, whether tuning is justified, a specific expected out-of-distribution failure and the limits of using the frozen set for repeated comparisons.	Multiple variables changed; no retraining; comparing different evaluation rows; hiding worse results; generic claims about generalization without a concrete failure case.

A lower win-rate or worse tuned output does not automatically fail C3; no improvement threshold was published for this stage. A valid comparison and justified conclusion are required. Keep judge identity and count strings visible: the ledger copies only numeric WEEK3_RESULTS values.

C4 | Model Optimization and Deployment Decisions

Module 4 - Optimized Analyst | Criteria 1/2

Competency: Explain parallelism and model-optimization options, compare compressed versions of the trained Analyst under a predeclared quality gate, and justify an accept, reject or fallback decision using measured quality, memory, latency, throughput and cost.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C4.1 Your optimization choice shows understanding	Technique note beside D4.1-D4.2; links to the selected model configurations.	Distinguish data from model parallelism, training from inference optimization, teacher-output from logit distillation, adapter folding from multi-model merging, and reduced-precision quantization. Identify the RiskLens bottleneck and justify the chosen route and trade-offs. Independent multi-GPU training, distillation and multi-parent merging are optional extensions, not required implementations.	Calling device placement data-parallel training; confusing adapter folding with merging several fine-tunes; claiming compression always accelerates inference; listing terms without a choice.
C4.2 Your quality gate precedes the comparison MUST PASS	D4.1 GATE and pre-score snapshot/version; baseline and candidate inventory.	Before compressed-model re-scoring, record justified maximum format and content drops, their units, the Stage 3 reference, candidate arms, hourly price and intended ship arm. Preserve the dated version; keep thresholds unchanged for that comparison. Explain why the tolerances are acceptable for the stated use case rather than choosing them to guarantee a pass.	Setting thresholds after seeing results; confusing percent with percentage points; changing the intended arm without a separate decision record; unjustified permissive tolerances.
C4.3 Your compression comparison uses the trained Analyst	D4.2 memory table; D4.3 saved generations, quality table and Module 4 checkpoint.	Produce at least two compression arms of the learner-trained Analyst and record memory footprints and ratios against half precision. Re-score each arm and the Stage 3 reference on the same frozen rows for format and content, with consistent settings. Show threshold comparisons and every verdict. Identify content cosine similarity as a proxy, not factual correctness.	Quantizing an untuned base; counting a configuration as a measured arm; measuring format only; different rows per arm; presenting cosine similarity as a factuality guarantee.

Keep the evidence locations, saved outputs and version records visible in the submitted notebook. The same criteria appear in the module rubric.

C4 | Model Optimization and Deployment Decisions

Module 4 - Optimized Analyst | Criteria 2/2

All criteria in this competency must be Met. C4.2 is the MUST PASS review gate.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C4.4 Your serving estimates are measured and qualified	D4.4 benchmark outputs for reference and each arm; settings and cost calculation.	Report p50/p95 latency, actual generated-token throughput and estimated cost per 1,000 reports for the reference and each arm. State hardware, run count, warm-up, input/output lengths and hourly-price/utilization assumptions. Label small-sample percentiles and cost estimates appropriately; explain why lower precision may save memory without reducing latency.	Using max_new_tokens as generated length; comparing unequal workloads without disclosure; calling notebook p95 a production SLA; treating the assumed GPU price as a verified bill.
C4.5 Your deployment decision respects the gate	D4.3 decision note; accepted/rejected arm inventory; fallback and limits.	Select a candidate that passes the declared gate or reject failed candidates and document a feasible fallback. Link the decision to quality, memory and runtime evidence. An honestly evaluated failed compression can meet this competency; it cannot be labeled production-ready. Keep the original gate record and distinguish any later candidate or fallback evaluation.	Declaring ship after a failed gate; hiding a quality drop behind a smaller file; silently changing the model; rewriting the threshold to make a candidate pass.

At Module 4, add a standalone checkpoint below your workspaces for the frozen-set format/content comparison, benchmarks and verdict. Save these outputs without calling analyze_product(). Run the starter's combined Stage 4 evaluation after integration in Module 5. Do not edit GIVEN functions or use stubs to satisfy the checkpoint.

C5 | Integrated System Delivery and Defense

Module 5 - Integrated RiskLens and Final Delivery | Criteria 1/2

Competency: Integrate the learner-built inspector, evidence engine and selected Analyst into a reproducible RiskLens pipeline, evaluate its behavior and failures across products, and explain its architecture, model choice and measurement limits using the submitted evidence.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C5.1 Your real components work together MUST PASS	D4.5 analyze_product(); three saved product demonstrations; selected Analyst record.	Compose the learner inspector, retriever and selected Analyst through the published contracts. Preserve the GOOD short-circuit and return note, evidence and report. Save at least three product demonstrations: one good and two defective, with traceable inputs and outputs. Use real components, not hard-coded substitutes. Label any fallback or failed-gate demonstration and its limits.	Demo stubs standing in for components; a JSON note passed where training used prose; missing good-product behavior; reports with no traceable evidence; a rejected model labeled ready.
C5.2 Your system evaluation locates failures	D4.7 batch outputs; timing table; product-level failure taxonomy.	Run the full pipeline on at least ten products and save end-to-end p50/p95 plus inspector, retrieval and Analyst timing. For every wrong or weak report, identify the responsible stage and evidence for that diagnosis; explicitly report if none are observed. Explain what the small workload cannot establish about production reliability or generalization.	Benchmarking only the Analyst; no stage timing; vague failure categories; silently skipping failures; claiming a small batch establishes production reliability.
C5.3 Your delivery is reproducible and documented	D4.6 deployment note; final run/load and show_ledger(); artifacts, environment and reflection.	Provide a clean top-to-bottom run/load with saved evaluation outputs, accessible model/data artifacts and environment/configuration details. Display the complete metrics ledger and preserve earlier comparisons separately. Document runtime choice, the bottleneck at 100 times traffic and a monitoring signal. Include a reflection of at most 300 words on trade-offs, trusted/uncertain numbers and next steps.	Missing weights or data; a notebook that depends on hidden state; outputs cleared; assuming overwritten ledger entries preserve all experiments; deployment claims without a runnable route.

Keep the evidence locations, saved outputs and version records visible in the submitted notebook. The same criteria appear in the module rubric.

C5 | Integrated System Delivery and Defense

Module 5 - Integrated RiskLens and Final Delivery | Criteria 2/2

All criteria in this competency must be Met. C5.1 is the MUST PASS review gate.

Criterion	Where your evidence lives	What earns "Met"	Common mistakes
C5.4 You can explain the submitted system	Explanation cells in final notebook; traceable code/outputs; exam evidence if separately issued.	Explain what one actual component does, why its design fits RiskLens and what would break if a named part were removed. Trace one demonstrated product through the applicable stages. Defend one reported metric and discuss a failed experiment or unsupported claim using saved evidence. Submit a written walkthrough; separately issued exam evidence may replace equivalent parts.	Reading code without explaining behavior; a generic architecture unrelated to the notebook; no supporting measurement; avoiding a failure or limitation; substituting polish for understanding.

Use a passing compressed arm when available. If all compressed candidates fail, a clearly labeled feasible reference-model fallback may demonstrate integration; it does not establish successful compression. Preserve the rejected-candidate evidence and the original gate. No paid hosting or separate recording is required.